

EVALUATION OF ACUTE AND CHRONIC HEPATOTOXIC EFFECT OF CMF IN BREAST CANCER PATIENTS

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ABSTRACT

CMF regimen contains Cyclophosphamide, Methotrexate and 5-fluorouracil which was developed in "Istituto Nazionale Tumori in Milan and has found to be effective in both pre-menopausal and post-menopausal breast cancer patients. Since it contains some hepatotoxic drug components so we tried to evaluate the cumulative hepatotoxicity of this combination, both in acute and chronic phases. In case of acute toxicity the regimen is found to be safe. However in case of chronic toxicity, it is found to produce mild effects in patients receiving less than three cycles and moderate effects in patients receiving more than three cycles. This represents that benefit versus risk should be assessed before giving CMF regimen to the patients, especially when other drugs like analgesics or antidepressants are prescribed with it.

INTRODUCTION

According to consensus development conference, held at the National Cancer Institute in Bethesda in Sept-1985, the CMF regimen has reduced 25% mortality, as adjuvant chemotherapy in breast cancer patients¹. No other adjuvant treatment has been able to achieve survival rates 12 years after mastectomy which are comparable to this regimen (Bonadonna and Robustelli-della).

MATERIALS AND METHODS

Phase I- Evaluation of Toxicities:

Six patients were included in this phase of study. They were female of biopsy proven breast cancer. They were receiving CMF as adjuvant to surgery and/or radiotherapy. Blood sample was collected from each patient. The CMF was administered in these patients having following doses of different component drugs.

- a) Cyclophosphamide 1000 mg
- b) Methotrexate 50 mg
- c) 5-fluorouracil 750 mg

All these drugs were given I. V. drip tube of Dextrose saline or normal saline. The second blood sample was collected after the completion of therapy.

Phase II - Evaluation of chronic toxicities:

Twenty-two patients were recruited in this part of study. They were female of biopsy proven breast cancer. They were newly diagnosed or have been received three or six cycles of CMF in the following dosage schedule:

- a) Cyclophosphamide 1000 mg
- b) Methotrexate 50 mg per three weeks
- c) 5-Fluorouracil 75 mg

All the drugs were given through I.V. drip tube of Dextrose saline or normal saline. Blood sample was collected for each patient. Seven healthy female volunteers were also

Table -1
Acute Hepatic Toxicities in CMF Treated Patients (N=6)

Hepatic Parameters	Before CMF	After CMF
Bilirubin (mg/dl)		
Total	02.91 ± 0.140 ⁺	02.67 ± 0.10
Direct	02.75 ± 0.140	02.59 ± 0.10
Transaminases (μ/ml)		
SGPT	36.33 ± 4.070	32.50 ± 5.25
SOOT	33.33 ± 2.810	37.50 ± 5.25
ALP(μ/l)	180.91 ± 48.52	186.16 ± 38.0
LDH(B-B μ/ml)	113.33 ± 21.51	31.66 ± 7.25*

⁺Mean ± S.E., *P<0.05

Table-2
Acute Hepatic Toxicities of Controls (n=7) and Patients Receiving less than three CMF (n=7)

Hepatic Parameters	Controls	Patients receiving less than three CMF
Bilirubin (mg/dl)		
Total	2.91 ± 0.13 ⁺	2.644 ± 0.11
Direct	2.52 ± 0.04	2.50 ± 0.12
Transaminases (μ/ml)		
SOOT	28.90 ± 3.50	39.20 ± 4.70*
SGPT	22.14 ± 2.96	31.16 ± 3.62*
ALP(μ/l)	118.28 ± 3.50	134.16 ± 23.53
LDH (B-B μ/ml)	39.28 ± 7.73	145.71 ± 32.06
Uric Acid (m dl)	4.89 ± 0.39	4.78 ± 0.47

⁺Mean ± S.E., *P<0.05

recruited, who were free from clinically demonstrable hepatic dysfunction and were using no pharmacotherapy of adverse hepatic effects. 5 cc-blood was taken from each control and dealt in the same manner as in case of patients.

Liver function was assessed by Bilirubin, Transaminases, Alkaline Phosphatase, LDH and Uric Acid. These parameters were determined by Merck Kits. Protein was not considered to be an important parameter due to interferences of administered drugs in protein metabolism

RESULTS

Table-1 shows important hepatic parameters just before and after the administration of regimen. No significant change ($P>0.05$) is noted in most of the parameters, except a decrease ($P<0.05$) in LDH after drug administration.

Table-2 shows same hepatic parameters in controls and in patients receiving less than three cycles of CMF. A statistically significant difference ($P<0.05$) is found in SOOT, SGPT and LDH.

Table-3 depicts hepatic parameters in controls and in patients receiving more than three cycles of CMF. A statistically significant difference ($P<0.05$) is present in ALP and LDH. However important changes although statistically insignificant ($P>0.05$) are observed in Bilirubin (both total and direct), SGPT and Uric acid.

DISCUSSION

A major advance in cancer chemotherapy during the last two decades is the use of

combination of drugs. This assertion is extensively supported by extensive experimental (Gldin *et al.*, 1974) and clinical (De-Vita and Young, 1975) evidences. For a combination which is designed on pharmacological basis, it is considered to contain the component drugs, which do not produce the same toxic effects.

However for a combination which is derived on clinical trials with prime importance of its response to particular malignancy, this important fact is sometime overlooked.

Such combinations sometimes contain the components having same loci of toxicities, e.g. CMF. It contains three component drugs which are all hepatotoxic, viz

1) Cyclophosphamide:

It is converted by the mixed hepatic function oxidase system to active metabolites (Connors *et al.*, 1974) and therefore hepatocytes are exposed to higher concentration of the toxic end products viz Phosphoraniide, mustard and acrolein (Cohnn

Table-3
Acute Hepatic Toxicities of Controls (n=7) and Patients Receiving more than Three CMF (n=7)

Hepatic Parameters	Before CMF	Patients receiving more than three CMF
Bilirubin (mg/dl)		
Total	$2.91 \pm 0.13^+$	3.24 ± 0.14
Direct	2.52 ± 0.04	3.13 ± 0.13
Transaminases (μ /ml)		
SGOT	28.90 ± 3.50	30.00 ± 4.26
SGPT	22.14 ± 2.96	28.57 ± 2.41
ALP (μ /l)	180.28 ± 3.50	$189.05 \pm 35.76^*$
LDH (B-B μ /ml)	39.28 ± 7.730	$120.00 \pm 25.99^*$
Uric Acid (mg/dl)	4.89 ± 0.39	4.20 ± 00.58

⁺Mean $\pm$ S.E., * $P<0.05$

et al., 1973). Perhaps higher activity of aldehyde oxidase of hepatic tissue provides safety from these products Deitrich, 1966 and Cox *et al.*, 1973). However hepatotoxicity of this drug has been reported (Calabresi and Chabner 1991).

2) Methotrexate:

Autoradiographic studies in mice (Darzynkiewicz *et al.*, 1966) and aminopterin labelled with a gamma emitter [1]-3-iodoaminopterin studies in human patients (Johns *et al.*, 1968) reveal its concentration in liver. Hepatotoxicity of transient elevations of hepatic enzymes have been noted with high dose infusions and severe hepatic pathology including cirrhosis has been found in low dosage therapy (Dahl *et al.*, 1971 and Roerig *et al.*, 1971).

3) 5-Fluorouracil:

Although hepatotoxicity of this drug has not been noted as yet, but metabolic degradation of the drug occurs in liver (Calabresi and Parks, 1985) which may enhance the toxicities of other drugs or may produce toxic substances after interaction with other metabolites of combination drugs.

In our studies the acute toxicities of these drugs when given simultaneously do not present an establish picture of toxic hepatic pathology but the ultrachanges of different hepatic parameters show that these changes may lead to cirrhosis (Kumar *et al.*, 1994) if persist for a long period or exaggerate on subsequent therapies or combinations. One may expect that these changes may easily be exaggerated if these patients are given hepatic enzyme inducing drugs especially neuroleptics or analgesics (Benet *et al.*, 1991) as they are usually prescribed to relieve pain or tension associated with a diagnosis of malignancy and depression linked with a loss of an important sexual part of the body as a result of mastectomy. The chronic toxicities as found by us do not depict, statistical significant alterations, in different parameters, however their mode is important with reference that cumulatively they show an

enzyme depletion phenomenon. However on summarizing the whole discussion, we can say that CMF does not produce considerable toxicity. This is in support to the studies done by Bonadonna (1988) on CMF toxicity. However this crucial question remains to be interrogated that is this combination is safe to combine with the supportive drugs which are used in addition to this regimen to relieve different symptoms? Theoretically keeping the results of our study in mind, our answer is "no". However the benefit versus risk should be considered in such cases.

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